# Olive Leaf Disease Detection Using Deep Learning

Empowering Palestinian Agriculture with AI

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This paper was prepared to document the development, implementation, and evaluation of an AI-based deep learning system for the early detection of olive leaf diseases, with the goal of supporting sustainable agriculture in Palestine.

## Abstract

This paper presents a deep learning approach for detecting and classifying olive leaf diseases using a transfer learning model based on MobileNetV2. To address the challenges of class imbalance and limited data, we incorporate advanced data augmentation, class weighting, and test-time augmentation (TTA). The model achieves a high classification performance with **92%** accuracy and a macro-averaged ROC-AUC score of **0.9867**. The system is designed to assist Palestinian farmers by providing a lightweight, interpretable, and accessible solution for early disease detection. A lightweight web application was developed to enable practical, real-time usage in field settings. This work highlights the potential of AI to enhance sustainable agriculture through intelligent disease monitoring and diagnosis.

## Nomenclature

| Abbreviation / Term | Definition |
|---|---|
| **Class Weighting** | A training technique used to handle imbalanced datasets by giving more importance to underrepresented classes |
| **CNN** | Convolutional Neural Network |
| **Feature Map** | The activation outputs of convolutional layers capturing spatial patterns |
| **ImageDataGenerator** | A Keras utility used for real-time image data augmentation |
| **LeakyReLU** | A type of activation function allowing small gradients when the unit is not active |
| **MobileNetV2** | A lightweight CNN architecture designed for mobile and embedded vision applications |
| **PR Curve** | Precision-Recall Curve |
| **ROC-AUC** | Receiver Operating Characteristic – Area Under Curve |
| **Transfer Learning** | A technique where a pre-trained model is adapted to a new, but related task |
| **TTA** | Test-Time Augmentation |

# Introduction

## a. Overview

Olive cultivation is a cornerstone of agricultural and economic life in many Mediterranean regions, particularly in Palestine. Diseases affecting olive trees—such as *aculus olearius* and *peacock spot*—can cause substantial damage, leading to reduced yield, compromised fruit quality, and economic losses. Early detection is critical to minimizing these impacts and supporting sustainable agriculture. However, traditional disease identification methods, which rely on human visual inspection, are prone to error, resource-intensive, and often inaccessible to farmers in remote or under-resourced areas.

Recent advances in Artificial Intelligence (AI), particularly in deep learning and computer vision, have enabled automated image-based plant disease detection. These methods allow farmers and researchers to detect diseases quickly and accurately from leaf images, enhancing the potential for timely intervention. This study explores the use of deep learning—specifically, transfer learning with the MobileNetV2 architecture—to classify olive leaf diseases from images. By integrating data augmentation, class weighting, and performance visualization, the model aims to provide reliable predictions, even in cases of class imbalance and limited data.

## b. Existing Problem

Traditional disease detection techniques in agriculture suffer from several practical limitations:

- Manual inspection is subjective and time-consuming, often leading to inconsistent or delayed responses.
- Expertise is not always locally available, especially in rural or low-income areas.
- Disease symptoms can overlap visually, making accurate diagnosis difficult without laboratory analysis.
- Datasets for specific plant species or local diseases are often scarce, reducing the effectiveness of general-purpose AI models.

Additionally, models trained without considering real-world constraints—such as class imbalance or field variation—may perform well in testing but fail under actual deployment scenarios. These issues highlight the need for an intelligent, scalable, and domain-specific solution for olive leaf disease detection.

## c. Related Work (Literature Survey)

Numerous studies have explored AI-based solutions for plant disease classification. Mohanty et al. (2016) used convolutional neural networks (CNNs) to classify 26 crop diseases, achieving over 99% accuracy on the PlantVillage dataset. However, that dataset was collected under controlled conditions, limiting real-world applicability.

MobileNet and its improved version, MobileNetV2 (Sandler et al., 2018), have gained popularity for being lightweight and efficient, especially for embedded and mobile applications. These models have been adapted for various plant classification tasks, including tomato and grapevine disease detection, due to their balance of speed and accuracy.

However, challenges remain—particularly when dealing with real-field data, class imbalance, and visually similar disease symptoms. Approaches such as class weighting, data augmentation, and test-time augmentation (TTA) have been proposed to enhance performance in such conditions (Shorten & Khoshgoftaar, 2019; Cui et al., 2019).

## d. Hypothesis

We hypothesize that a transfer learning approach based on MobileNetV2, when combined with targeted data augmentation, class balancing, and model interpretability techniques, can effectively classify olive leaf diseases—even with a limited, imbalanced dataset—and generalize well to real-world agricultural images.

## e. Motivation for Carrying Out the Research (Objectives)

This research is driven by the need to:

- Build a lightweight yet accurate model for olive leaf disease classification.
- Address class imbalance and limited labeled data through smart augmentation and training strategies.
- Offer a practical solution that can be deployed via a web application for use by Palestinian farmers.
- Contribute to the body of knowledge on domain-specific deep learning applications in agriculture.

## f. Main Contribution

The main contributions of this study include:

- A MobileNetV2-based deep learning pipeline optimized for olive leaf disease classification.
- Application of **data augmentation**, **class weighting**, and **TTA** to address real-world constraints.
- Use of **visual diagnostics** such as feature maps, confusion matrices, and filter visualization to interpret model decisions.
- Deployment of the trained model as part of a lightweight **web application** interface to assist farmers in the field.
- A comprehensive evaluation using accuracy, ROC-AUC, precision-recall curves, and misclassification analysis.

## g. Report Organization

The remainder of this report is structured as follows:

- **Section 1** describes the dataset, preprocessing steps, model architecture, and training procedures.
- **Section 2** provides a literature review of AI applications in plant disease detection.
- **Section 3** presents a comparative view through related case studies.
- **Section 4** discusses experimental results, performance metrics, and visual analysis.
- **Section 5** concludes the paper and outlines possible directions for future work.

## Section 1: Methodology

This section outlines the complete pipeline used to develop the olive leaf disease classification system, including data sources, preprocessing strategies, model architecture, training setup, and evaluation metrics.

### a. Dataset Description

The dataset used in this study contains **3 classes** of olive leaf images:

- *aculus_olearius* (diseased)
- *peacock_spot* (diseased)
- *healthy*

The images were organized into `train` and `test` directories, with a **20% validation split** taken from the training set during data loading. The dataset consists of:

- 1,776 training images
- 444 validation images
- 680 test images

All images were resized to 224x224 and normalized using rescale=1./255. Data augmentation techniques included:

- Rotation: up to 30 degrees
- Shear, Zoom, Brightness, Width/Height shifts
- Horizontal & vertical flipping

The class distribution was imbalanced, with *peacock_spot* being the most represented and *aculus_olearius* the least. To mitigate this, we applied class weighting and data augmentation (see subsections below).

### b. Data Preprocessing and Augmentation

To increase data diversity and simulate real-world variability (e.g., lighting, orientation, size), we used the `ImageDataGenerator` class from Keras with the following augmentations:

- Rotation: ±30°
- Width/Height shift: ±10%
- Zoom range: up to ±30%
- Shear transformation and brightness variation (0.7 to 1.3)
- Horizontal and vertical flipping
- Fill mode: 'nearest'

Additionally, all images were:

- Resized to (224, 224)
- Normalized to the [0, 1] range via rescale=1./255
- Shuffled with fixed seeds to ensure reproducibility

Validation and test images were **not augmented**, only rescaled.

## c. Transfer Learning with MobileNetV2

We adopted **MobileNetV2** as the base model due to its lightweight architecture and strong performance in mobile vision tasks [1]. Key characteristics of the model setup:

- Pretrained on ImageNet
- Top (dense) layers removed
- Convolutional base retained

We **froze all BatchNormalization layers** during training to preserve stable feature statistics [2], while allowing other layers to fine-tune.

The classification head added to the model included:

- Global Average Pooling
- Dropout (rate 0.4 and 0.3)
- Dense layers (256 and 128 units) with **LeakyReLU** activations
- Final Dense layer with **softmax** activation for 3-class output

## d. Training Strategies and Hyperparameters

Training was performed using the following settings:

- **Optimizer:** Adam
- **Learning rate:** 1e-4 (with ReduceLROnPlateau callback)
- **Loss function:** Categorical Crossentropy
- **Batch size:** 32
- **Epochs:** up to 30
- **EarlyStopping:** patience = 5
- **ModelCheckpoint:** save best model based on validation loss
- **Random seed:** 42 (for reproducibility)

Class imbalance was addressed using **`compute_class_weight()`** from `sklearn`, which increased the penalty for underrepresented classes during loss calculation.

## e. Evaluation Metrics and Class Weighting

To evaluate model performance comprehensively, we tracked:

- Accuracy , Precision, Recall, F1-score
- Top-2 Accuracy, ROC-AUC (macro and per class)
- Confusion Matrix, Precision-Recall Curves

For robustness, we also applied **TTA** during inference by averaging predictions over 5 augmented views of each image.

# Section 2: Literature Review

This section provides a background on the application of Artificial Intelligence in agriculture, with a focus on plant disease detection methods, recent AI-based approaches, and the challenges they address.

## a. AI in Agriculture

Artificial Intelligence (AI) has emerged as a transformative force in agriculture, enabling smarter decision-making, automation, and predictive analytics. Applications range from crop yield forecasting and soil monitoring to pest control and disease diagnosis. Among these, **computer vision for plant disease detection** is one of the most practical and impactful areas, especially in regions with limited access to agricultural experts.

AI helps bridge the gap between knowledge and accessibility by using machine learning algorithms to analyze patterns in leaf color, shape, and texture—features that often correlate with specific diseases. This facilitates **early-stage detection**, which is critical for effective disease management.

## b. Plant Disease Detection Methods

### 1. Traditional Methods

Traditional approaches to plant disease identification rely on:

- Visual inspection by trained agronomists
- Laboratory testing for pathogen identification
- Expert system checklists or rule-based systems

While these methods can be accurate, they are often **slow, subjective, and costly**. They are also **inaccessible** in remote regions where expert knowledge and lab resources are not available.

### 2. AI-Based Approaches

Deep learning, especially **Convolutional Neural Networks (CNNs)**, has been widely adopted for plant disease classification due to their ability to extract spatial hierarchies in image data. Early work by Mohanty et al. (2016) demonstrated that deep CNNs could classify over 25 plant diseases with **over 99% accuracy** on the PlantVillage dataset [1].

Since then, various architectures have been tested, including:

- VGG16 / VGG19
- ResNet50 / ResNet101
- InceptionV3
- MobileNet / MobileNetV2

MobileNetV2 [2], in particular, is known for its **efficiency and speed**, making it well-suited for deployment on mobile and edge devices.

## c. Applications of AI in Plant Disease Detection

AI-powered plant disease detection has been successfully applied to:

- Tomato leaf diseases (e.g., blight, mosaic virus)
- Banana and cassava plant health
- Apple and grapevine leaf disorders
- Rice blast and bacterial leaf blight

These applications show that CNNs can generalize well across different crops and regions when supported by **adequate preprocessing and augmentation**. However, many models are tested only on ideal datasets and do not generalize to **real-world agricultural settings**, where noise, lighting, and class imbalance are major issues.

### d. Integration with IoT

Recent developments include the integration of AI models with **Internet of Things (IoT)** systems for real-time monitoring. These systems use connected sensors and cameras to capture leaf images and process them through AI models, either on-device or via the cloud. Such pipelines support **early warning systems**, enabling farmers to act before diseases spread.

In low-resource settings, however, simpler **web-based or mobile deployment** may be more realistic. Lightweight models like MobileNetV2 make this feasible, especially when coupled with test-time augmentation and visual interpretability features.

### e. Limitations and Challenges

Despite significant progress, several challenges remain:

- **Class imbalance**: Common in datasets where some diseases are underrepresented.
- **Limited labeled data**: Especially for niche or region-specific diseases.
- **Visual similarity**: Different diseases may appear similar to both humans and AI models.
- **Field variability**: Changing lighting, angles, and background noise reduce model accuracy.
- **Lack of interpretability**: Many CNNs are black boxes, which reduces trust in their predictions.

This study directly addresses these limitations through a carefully designed training pipeline with **augmentation, class balancing, and visual interpretability tools**.

## Section 3: Case Studies and Comparative Applications

This section presents a focused case study on olive leaf disease detection using deep learning, followed by a brief comparison with related systems developed for other plant species such as tomato and chili.

### a. Olive Leaf Disease Detection with MobileNetV2 (This Study)

The core of this research is a MobileNetV2-based deep learning system that classifies olive leaf images into three categories: *aculus_olearius*, *peacock_spot*, and *healthy*. The model was trained using a relatively small and imbalanced dataset under real-world conditions, making generalization a primary challenge.

Key innovations in this case study include:

- **Transfer Learning:** Leveraging pretrained ImageNet weights to compensate for limited olive-specific training data.
- **Class Weighting:** Addressing imbalance using computed class weights in the loss function.
- **Data Augmentation:** Simulating real-world conditions via rotation, brightness, zoom, flip, and shear.
- **BatchNorm Freezing:** Stabilizing fine-tuning by freezing BatchNormalization layers.
- **TTA:** Averaging predictions over multiple augmented views for higher robustness.
- **Interpretability Tools:** Including confusion matrices, ROC curves, filter visualizations, and misclassification analysis.

#### Final Results

- **Accuracy:** 92%
- **Macro ROC-AUC:** 0.987
- **Precision-Recall (Class-level):** F1-scores above 0.90 for all classes

### b. Tomato Leaf Disease Detection (Review)

Tomato plants are a popular subject in AI plant pathology research due to the availability of large annotated datasets like PlantVillage. CNNs such as ResNet50 and InceptionV3 have achieved over 95% accuracy on tomato leaf disease classification under controlled settings.

However, most of these models rely on **clean, studio-lit images** that fail to generalize in field conditions. While MobileNet has also been applied successfully to tomatoes, its performance drops when tested on real farm data, indicating a gap that this study addresses.

### c. Chili Plant Disease Detection with CNNs (Review)

Recent work on chili plants has used shallow CNNs for disease identification with moderate success. These models tend to overfit when trained on small datasets, highlighting the importance of **transfer learning** and **strong augmentation**strategies—both of which are integrated into our olive leaf disease model.

Moreover, these approaches often lack interpretability features, such as feature map visualization or misclassification analysis, which are key components of our pipeline to ensure trust and usability for end-users like farmers.

## Section 4: Results and Discussion

This section presents a comprehensive analysis of the model's performance based on quantitative metrics, visual diagnostics, and error analysis. The evaluation is conducted on a held-out test set of 680 olive leaf images.

## a. Model Training Performance (Loss/Accuracy Curves)

The model converged effectively within 16 epochs, demonstrating stable generalization with no significant overfitting. Both training and validation loss decreased consistently, while accuracy improved until early stopping was triggered.

### Training Curves

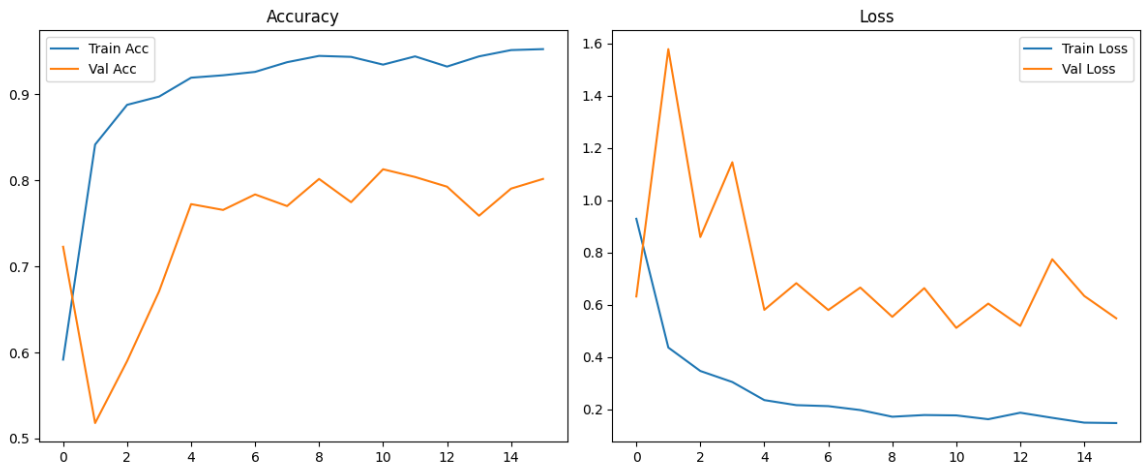


Figure 1: Accuracy and Loss curves

- **Best Validation Accuracy:** 0.92
- **Training stopped early** at epoch 16 to prevent overfitting
- Learning rate was reduced adaptively using `ReduceLROnPlateau`

## b. Confusion Matrix and Metrics

The final model achieved a **test accuracy of 92%**, with strong per-class performance:

### Classification Report

| Class | Precision | Recall | F1-score |
|---|---|---|---|
| **Aculus Olearius** | 0.97 | 0.83 | 0.90 |
| **Healthy** | 0.91 | 0.96 | 0.93 |
| **Peacock Spot** | 0.89 | 0.94 | 0.92 |
| **Overall** | 0.92 | 0.92 | 0.92 |

### Confusion Matrix

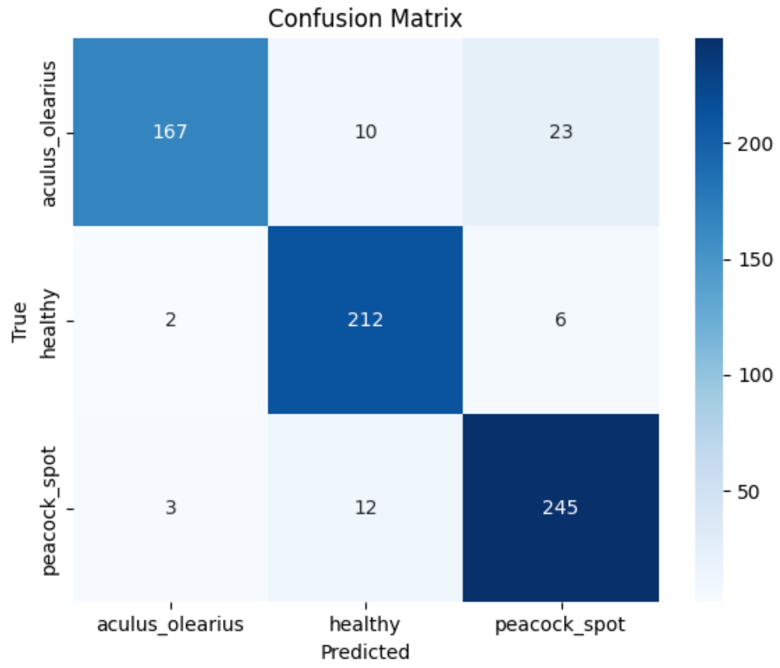


Figure 2: Confusion Matrix

Most errors were mild misclassifications between *peacock_spot* and *aculus_olearius*, which often exhibit visually similar symptoms such as dark spots and surface discoloration. This overlap makes distinguishing between these two diseases more challenging for the model.

## c. ROC and Precision-Recall Curves

ROC and PR curves provide a deeper look at class-wise confidence and separability.

### Per-Class ROC Curves

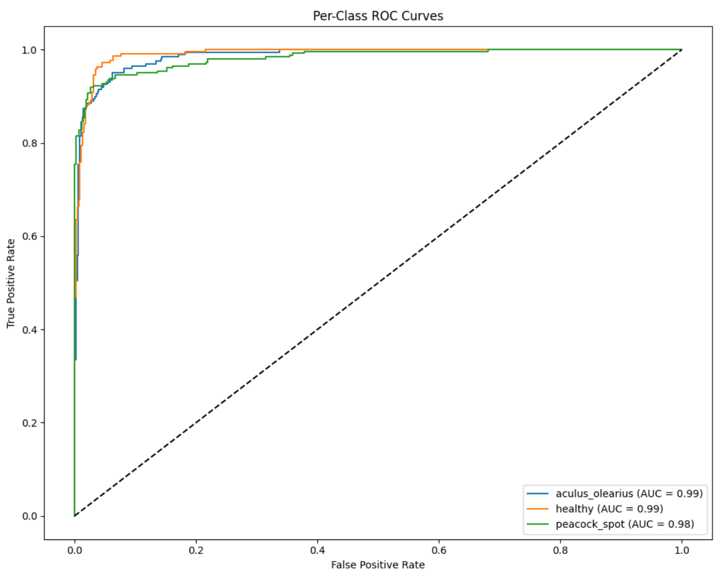


Figure 3: ROC Curve for Class-wise Performance

- **Macro-Averaged ROC-AUC**: 0.987
- All classes demonstrated excellent separability, with AUC values > 0.97

Precision-Recall Curves:

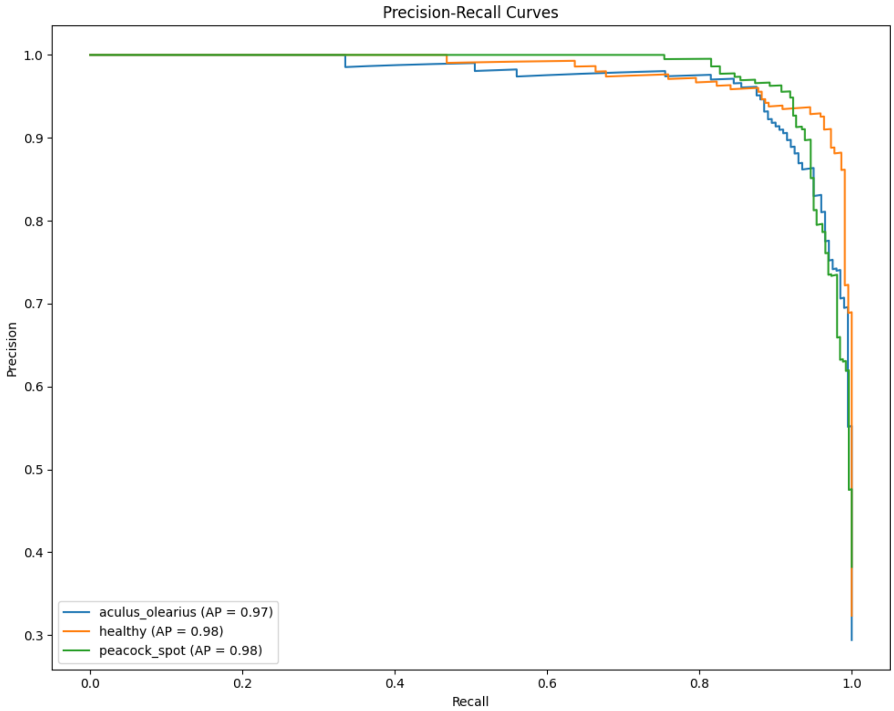


Figure 4: Precision-Recall Curves

- All average precision (**AP**) scores were > 0.95
- The model maintains high precision across a wide recall range

## d. Analysis of Misclassifications

### Misclassified Sample Visuals


Figure 5: Misclassified Sample Visuals

Misclassifications were mostly caused by:

- Visual similarity between *peacock_spot* and *aculus_olearius*, especially in early stages
- Leaf image artifacts, such as spot clustering, glare, or reflections, leading to ambiguity
- Augmentation noise, where certain transformations (e.g., heavy shear or zoom) slightly distorted key features, even with TTA applied

## e. Model Limitations and Interpretability

### Feature Maps (top_conv)

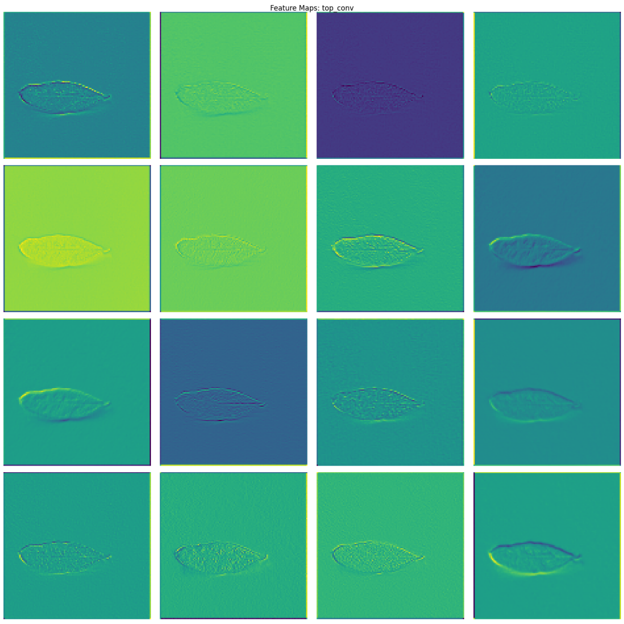


Figure 6: Feature Maps (top_conv)

Activation maps demonstrate that the network focuses primarily on leaf structure features, minimizing background influence.

### Class Distribution Analysis

The predicted class distribution closely mirrors the true labels, indicating no dominant class bias.

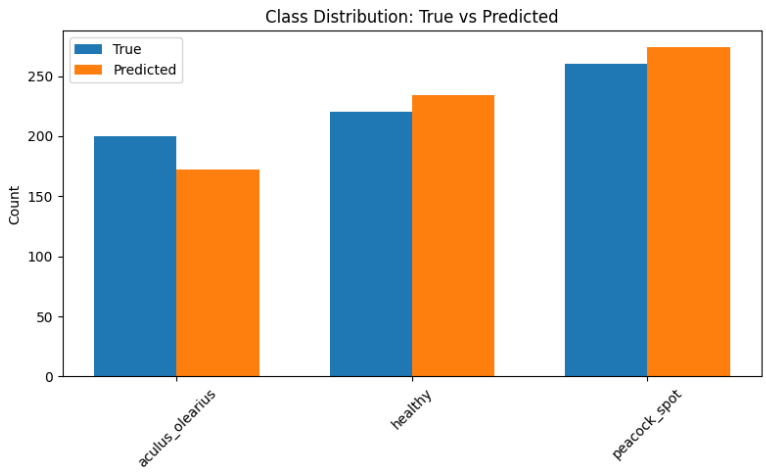




Figure 7: Class Distribution Analysis

### f. Summary of Findings

- MobileNetV2 with class weighting and TTA can generalize well to real-world olive leaf disease images.
- The model achieved **92% test accuracy**, with **macro ROC-AUC = 0.987**.
- Visual interpretability features validate that the model focuses on meaningful regions.
- Misclassifications are limited and mostly understandable under noisy field conditions.

## Section 5: Conclusion and Future Work

### a. Conclusion

This study presented a deep learning-based system for olive leaf disease classification using MobileNetV2 and transfer learning. The pipeline was designed to handle class imbalance and limited data by integrating advanced data augmentation, class weighting, BatchNorm freezing, and test-time augmentation (TTA).

The final model demonstrated:

- **92% test accuracy**
- **Macro-averaged ROC-AUC of 0.987**
- Strong per-class precision, recall, and F1-scores
- Robustness to visual variability via extensive augmentation
- Interpretability through feature maps, and misclassification analysis

By deploying a lightweight, interpretable model with high diagnostic accuracy, this research shows the potential of AI to support sustainable agriculture, especially in under-resourced areas such as rural Palestine.

### b. Implications for Agriculture

The system developed here can serve as a foundation for:

- **Early warning systems** for olive diseases
- **Farmer-facing mobile or web applications**
- **Automated monitoring** in IoT-enabled agricultural platforms
- **Data-driven interventions** for improving crop health and yield

In low-infrastructure environments, a solution like this offers a **cost-effective, scalable tool** to bridge the gap between expert diagnostics and real-world practice.

### c. Recommendations for Future Work

While this study achieved strong results, several enhancements can be explored in future work:

1. **Larger, more diverse datasets:** including seasonal and geographic variations.
2. **Explainable AI (XAI) techniques:** such as Grad-CAM or SHAP to enhance trust.
3. **Multi-disease detection:** across more olive leaf conditions and fruit health.

4. **Mobile deployment:** converting the model to TensorFlow Lite for on-device inference.
5. **Real-time feedback systems:** integrating the model into an interactive decision-support app for farmers.

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